EXPERIMENT-2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

BRAIN TEST – UI/UX DESIGN USING CHUNKING PRINCIPLE

Objective

To design a **Brain Test application** that evaluates a user's **visual memory** by applying the **chunking principle**, where information is grouped into meaningful units to improve recall and reduce cognitive load.

FRAME 1: INSTRUCTION PAGE

Purpose

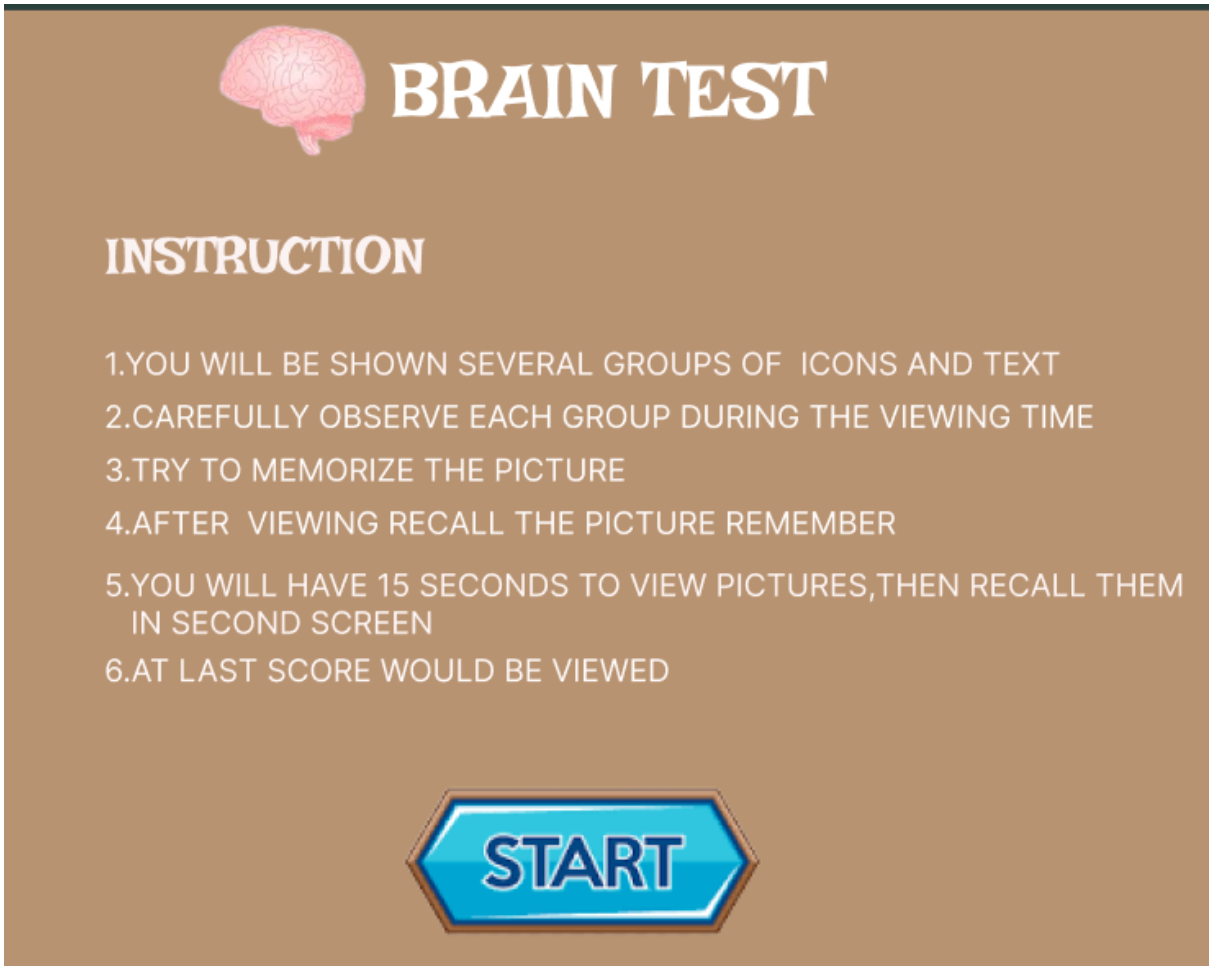
- Introduces the task clearly before the test begins.
- Prepares users mentally for the memory challenge.

UI & UX Analysis

- **Title "BRAIN TEST"** with a brain icon immediately communicates the purpose of the application.
- Instructions are presented as **numbered steps**, which is an example of **textual chunking**.
- Each instruction conveys **one action only**, avoiding overload.
- The **15-second time limit** is clearly mentioned, setting expectations.
- The **START button** is visually highlighted, guiding the user's next action.

Chunking Benefits

- Instructions are broken into **small, digestible steps**.
- Users process information sequentially instead of all at once.
- Improves comprehension and reduces confusion.



FRAME 2: CHUNKING PHASE (MEMORIZATION SCREEN)

Purpose

- This is the **memory encoding phase** where users observe and memorize images.

UI Components

- **Countdown timer (15 seconds)** creates urgency and focus.
- **Progress bar** visually represents remaining time.
- **Grid of icons** arranged evenly in rows and columns.
- Icons belong to **familiar categories** such as:
  - Animals (cat, bird, duck)
  - Objects (house, bottle)
  - Nature (sun, snowflake)
  - Food (watermelon)
- Decorative brain and idea bulb visuals maintain engagement.

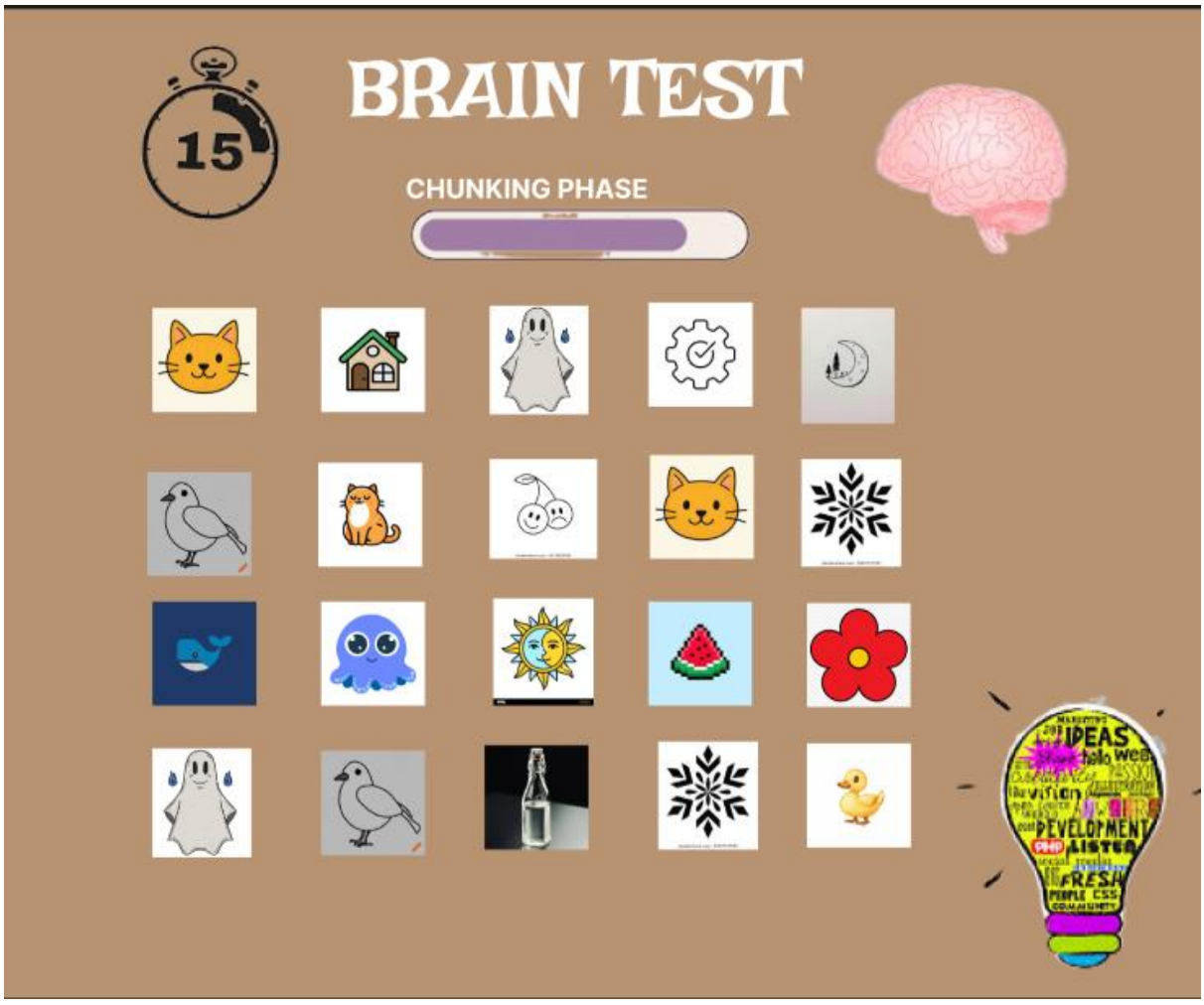
Chunking Strategy Used

- Icons are grouped visually in a **grid layout**, encouraging users to remember them as **sets instead of individuals**.
- Users subconsciously form mental groups based on **similar shapes, themes, or colours**.

Cognitive Benefit

- Enhances **short-term memory retention**
- Reduces cognitive overload
- Improves pattern recognition

FRAME 3: RECALL PHASE (SELECTION SCREEN)



Purpose

- **Tests the user's ability to retrieve stored visual information.**

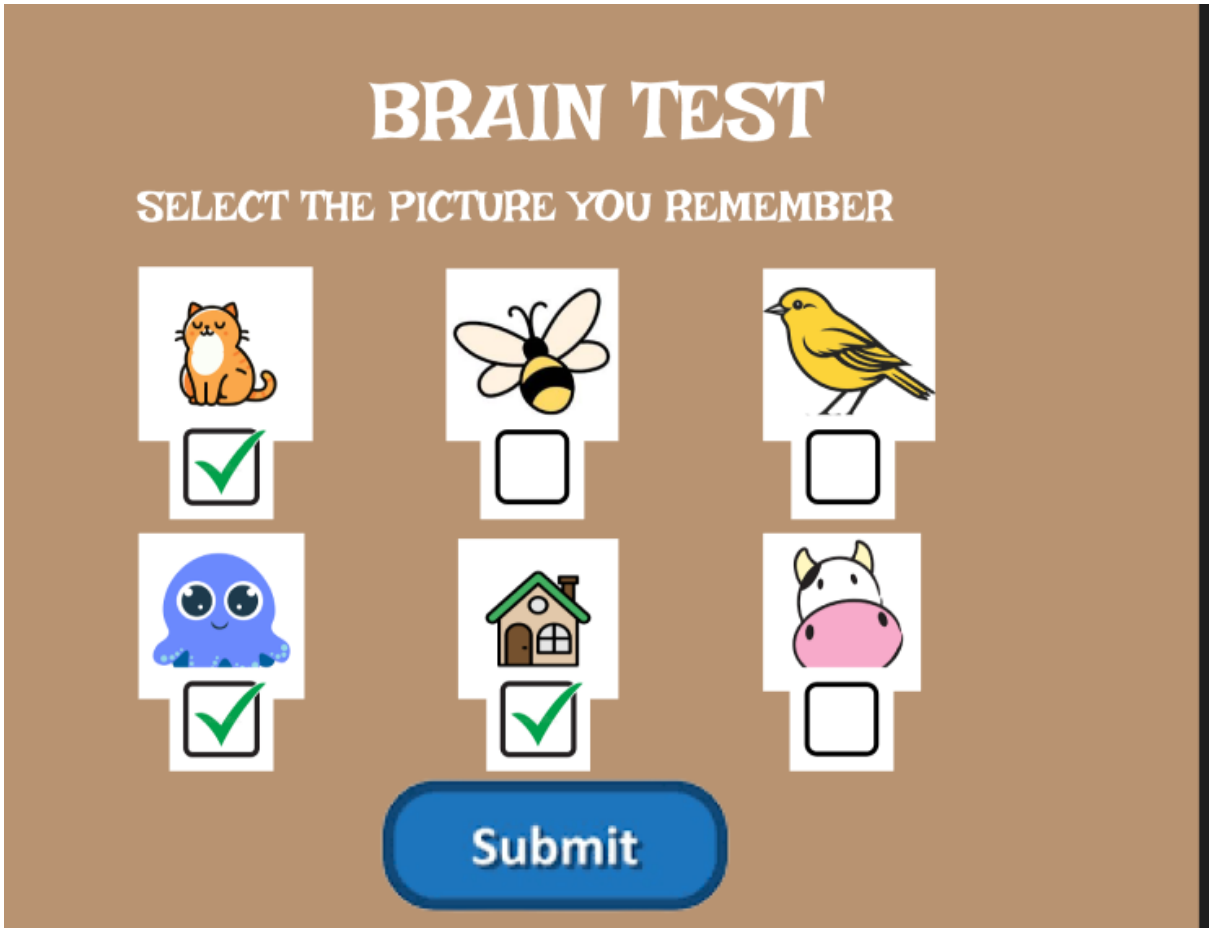
UI & Interaction

- **Clear instruction: "SELECT THE PICTURE YOU REMEMBER"**
- **Multiple image options are shown:**
  - **Some were present in the chunking phase**
  - **Some are distractor images**
- **Checkboxes allow users to make selections.**
- **Submit button confirms the recall response.**

UX Benefits

- **Simple selection mechanism reduces errors.**
- **Distractors test memory accuracy, not guessing.**
- **Visual consistency helps users focus only on recall.**

FRAME 4: RESULT PAGE



Purpose

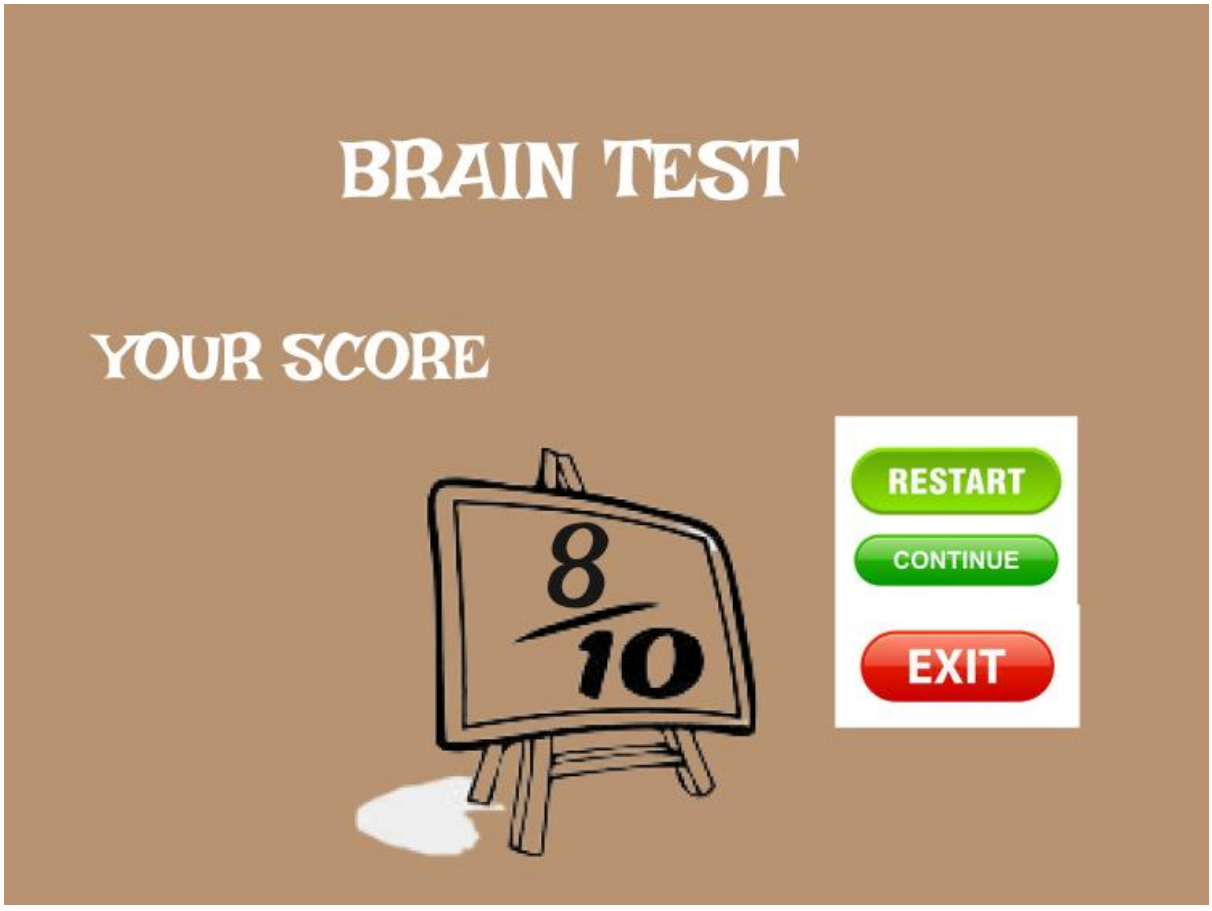
- Displays user performance and provides feedback.

UI Elements

- Score shown clearly as **8/10**
- Visual scoreboard enhances gamification.
- Three action buttons:
    - **Restart** – retake the test
    - **Continue** – proceed to next level
    - **Exit** – end the test

UX Benefits

- Immediate feedback improves learning.
- User control over next action increases engagement.
- Gamified visuals reduce stress and improve motivation.



PROTOTYPE LINK:

https://www.figma.com/design/iky0Bxn4juJUNUAo5fym9n/Untitled?node-id=0-1&m=dev&t=ufu6KwKwaJK3DT6A-1